

B.Tech. Degree IV Semester Regular/Supplementary Examination

June 2024

EE 19-209-0404 POWER ELECTRONICS
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Identify the switching characteristics various power semiconductor devices.
CO2: Analyse the characteristics of different uncontrolled and controlled rectifiers.
CO3: Design various inverter circuit topologies based on application and knowledge of PWM technique.
CO4: Analyse the working of different switched mode converters.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
L6 – Create

PO – Programme Outcome

PART A

(Answer ALL questions)

(8 × 3 = 24) Marks BL CO PO

- | | | | | | |
|--------|--|---|--------|---|---|
| I. (a) | Describe the basic structure and static characteristics of a power MOSFET. | 3 | L2 | 1 | 1 |
| (b) | Explain the difference between natural and forced commutation in thyristors. | 3 | L2 | 1 | 1 |
| (c) | Calculate the average output voltage of a single-phase fully controlled bridge rectifier for a firing angle of 30°. Assume the input voltage is 230 V RMS. | 3 | L3 | 2 | 1 |
| (d) | Explain the effect of source inductance on the performance of a three-phase fully controlled rectifier. | 3 | L2 | 2 | 1 |
| (e) | Explain the principle of sinusoidal PWM technique used in single-phase inverters. | 3 | L2 | 3 | 1 |
| (f) | Describe the working principle of a single-phase AC voltage controller with R load. | 3 | L2 | 3 | 1 |
| (g) | Explain the principle of operation of a buck-boost converter and derive the expression for its voltage transfer function. | 3 | L2, L3 | 4 | 1 |
| (h) | Describe the basic building blocks of an online UPS system. | 3 | L2 | 4 | 1 |

PART B

(4 × 12 = 48)

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|-----------|---|----|----|---|---|
| II. | Describe the basic structure and operation of a Silicon Controlled Rectifier (SCR). Explain its VI characteristics with a neat diagram. | 12 | L2 | 1 | 1 |
| OR | | | | | |
| III. | Explain the turn-on methods for an SCR. Describe the R and RC triggering circuits with neat diagrams. | 12 | L2 | 1 | 1 |

(P.T.O.)

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		Marks	BL	CO	PO
IV. /	Explain the operation of a single-phase half-wave controlled rectifier with R load. Draw the relevant waveforms for input voltage, output voltage and SCR current.	12	L2	2	1,2
OR					
V.	Describe the working principle of a three-phase full-wave uncontrolled rectifier. Draw the circuit diagram and output voltage waveform. Calculate the average output voltage if the input is 415 V (line-to-line).	12	L3	2	1,2
VI. /	Explain the operation of a single-phase full-bridge inverter with resistive load. Draw the circuit diagram and output voltage waveform for 180° conduction mode.	12	L2	3	1,2
OR					
VII. /	Describe the principle of sinusoidal PWM technique used in inverters. Draw the output voltage waveforms. List three advantages of this technique.	12	L2	3	1,2
VIII. /	Describe the working principle of a boost converter. Draw the circuit diagram and waveforms for continuous conduction mode. Calculate the duty cycle required to boost 12 V to 24 V.	12	L2,L3	4	1,2
OR					
IX.	Explain the operation of a buck converter with neat circuit diagram and waveforms. Derive the expression for its voltage transfer function.	12	L2,L3	4	1,2

Blooms's Taxonomy Level's

L2 - 73.68%, L3 - 26.32%,
